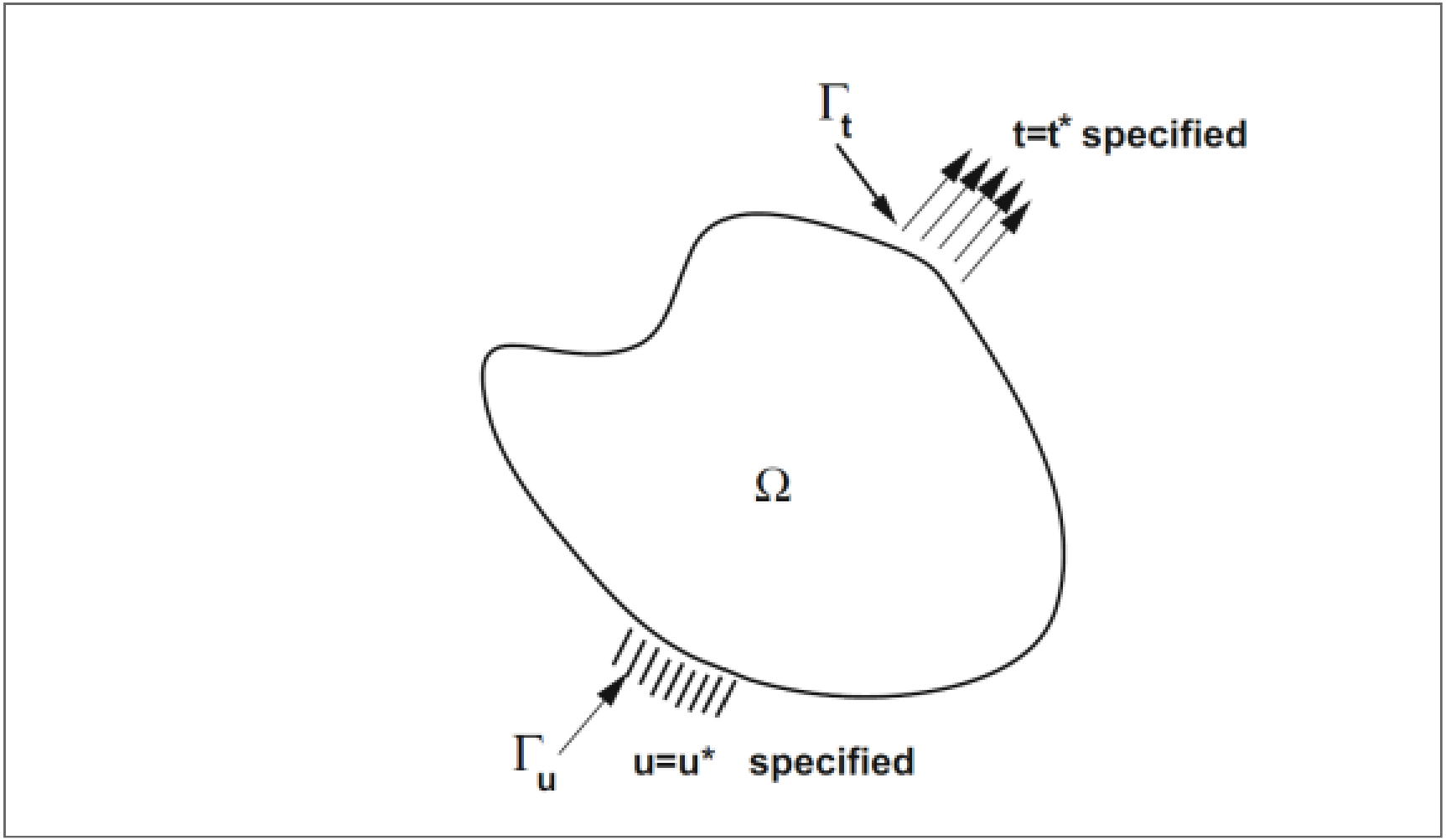
Quas-Static Elasticity in 3D

- Starting with:
  $\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = 0$
- We multiply by a test function $\mathbf{v}$ and integrate over the domain Ω. This is the method of weighted residuals, where $\boldsymbol{r}$ is the residual::
  $$\int_{\Omega} (\nabla \cdot \boldsymbol{\sigma} + \boldsymbol{f}) \cdot (v)\, d\Omega = 0 = \int_{\Omega} \boldsymbol{r} \cdot \boldsymbol{v} d\Omega$$



- Recall the product rule for the divergence of a tensor contracted with a vector::

$$\nabla \cdot (\boldsymbol{\sigma}\mathbf{v}) = (\nabla \cdot \boldsymbol{\sigma}) \cdot \mathbf{v} + \boldsymbol{\sigma} : \nabla\mathbf{v}$$

- we can write:
  $(\nabla \cdot \boldsymbol{\sigma}) \cdot \mathbf{v} = \nabla \cdot (\boldsymbol{\sigma}\mathbf{v}) - \boldsymbol{\sigma} : \nabla\mathbf{v}$
- Substituting this into our integrated equation and applying the divergence theorem:
  $$\int_\Omega \nabla \cdot (\boldsymbol{\sigma}\mathbf{v}) d\Omega - \int_\Omega \boldsymbol{\sigma} : \nabla\mathbf{v} d\Omega + \int_\Omega \mathbf{f} \cdot \mathbf{v} d\Omega = 0$$
  $$\implies \int_{\partial\Omega} (\boldsymbol{\sigma}\mathbf{v}) \cdot \mathbf{n} d\Gamma - \int_\Omega \boldsymbol{\sigma} : \nabla\mathbf{v} d\Omega + \int_\Omega \mathbf{f} \cdot \mathbf{v} d\Omega = 0$$
  The term $(\boldsymbol{\sigma}\boldsymbol{n})$ is the traction vector $\boldsymbol{t}$. The term $\nabla\boldsymbol{v}$ is the gradient of the test function, and its symmetric part is the strain tensor for the test function, $\boldsymbol{\varepsilon}(\boldsymbol{v})$.
- The boundary $\partial\Omega$ is composed of two parts:
- Γ_u (where displacements are prescribed)

- Γ_t (where tractions are prescribed).
- On Γ_u we enforce Dirichlet boundary conditions, meaning the displacement field $\mathbf{u}$ is prescribed as $\mathbf{u}^*$.
- On Γ_t we enforce Neumann boundary conditions, meaning the traction vector $\boldsymbol{\sigma n} = \boldsymbol{t}$ is prescribed as $\boldsymbol{t}^*$.
- This is the natural boundary condition that will be incorporated into the weak form. The boundary integral becomes:

$$\int_{\partial\Omega} \mathbf{t} \cdot \mathbf{v} d\Gamma = \int_{\Gamma_t} \mathbf{t}^* \cdot \mathbf{v} d\Gamma + \int_{\Gamma_u} \mathbf{t} \cdot \underbrace{\mathbf{v}}_{=0} d\Gamma = \int_{\Gamma_t} \mathbf{t}^* \cdot \mathbf{v} d\Gamma$$

We can now state our problem in its final weak form:
Find the displacement field $\boldsymbol{u} \in H^1(\Omega)$ such that $\boldsymbol{u} = \boldsymbol{u}^*$ on the Dirichlet boundary Γ_u for all test functions $\boldsymbol{v} \in H^1$ for

which $\boldsymbol{v}|_{\Gamma_u} = 0$ satisfying the weak form:

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega = \int_{\Omega} \mathbf{f} \cdot \mathbf{v} d\Omega + \int_{\Gamma_t} \mathbf{t}^* \cdot \mathbf{v} d\Gamma$$

Quas-Static Elasticity in 3D contd.

In the weak form

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega = \int_{\Omega} \mathbf{f} \cdot \mathbf{v} d\Omega + \int_{\Gamma_t} \mathbf{t}^* \cdot \mathbf{v} d\Gamma$$

we have the stress tensor $\boldsymbol{\sigma}$ expressed in terms of the strain tensor $\boldsymbol{\varepsilon}$ and the displacement field $\boldsymbol{u}$

We use the constitutive law for linear elasticity $\boldsymbol{\sigma} = \boldsymbol{L} : \boldsymbol{\varepsilon}$, where $\boldsymbol{L}$ is the fourth-order elasticity tensor. For an isotropic material, the matrix form of $\boldsymbol{L}$ (in Voigt notation)

is given by:

$$\mathbf{L} = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix}$$

where E is Young's modulus and v is Poisson's ratio. and $\boldsymbol{\sigma} = \boldsymbol{L} : \boldsymbol{\varepsilon}$ reffers to the double contraction of the elasticity tensor with the strain tensor, which can be expressed in index notation as:

$\sigma_{ij} = L_{ijkl}\varepsilon_{kl}$

where σ_{ij} are the components of the stress tensor, L_{ijkl} are the components of the elasticity tensor, and ε_{kl} are the components of the strain tensor.
in voigt notation, the strain tensor is represented as a vector:

$$\varepsilon = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{31} \\ 2\varepsilon_{12} \end{bmatrix}$$

and the stress tensor is represented as:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{31} \\ \sigma_{12} \end{bmatrix}$$

- We continue from our weak form:

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}), d\Omega = \int_{\Omega} \mathbf{f} \cdot \mathbf{v}, d\Omega + \int_{\Gamma_t} \mathbf{t}^* \cdot \mathbf{v}, d\Gamma$$

- Now, we substitute the constitutive law into the left-hand side of the equation:

$$\int_{\Omega} (\boldsymbol{L} \cdot \boldsymbol{\varepsilon}(\mathbf{u})) \cdot \boldsymbol{\varepsilon}(\mathbf{v}), d\Omega = \int_{\Omega} \mathbf{f} \cdot \mathbf{v}, d\Omega + \int_{\Gamma_t} \mathbf{t}^* \cdot \mathbf{v}, d\Gamma$$

Where we dropped the doble contraction (:) with $\cdot$ since we use the voigt notation for the strain and stress tensors. This equation has a standard structure. The left side is linear with respect to both the solution u and the test function v. The right side is linear only with respect to the test function v.

- We can define a bilinear form B(u,v) from the left-hand side, which represents the internal virtual work:

$$B(\mathbf{u}, \mathbf{v}) := \int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{v}) \cdot \boldsymbol{L} \cdot \boldsymbol{\varepsilon}(\mathbf{u}), d\Omega$$

- And we define a linear form F(v) from the right-hand side, which represents the external virtual work done by body

forces and applied tractions:

$$F(\mathbf{v}) := \int_{\Omega} \mathbf{f} \cdot \mathbf{v}, d\Omega + \int_{\Gamma_t} \mathbf{t}^* \cdot \mathbf{v}, d\Gamma$$

- The problem is now stated in its abstract form: Find the displacement u (satisfying Dirichlet BCs) such that for all valid test functions v:

  $$B(\mathbf{u}, \mathbf{v}) = F(\mathbf{v})$$

  This abstract form is the direct starting point for the Finite Element discretization, where we will replace the infinite-dimensional function spaces with finite-dimensional approximations.

- Recalling the definition of the strain tensor:

  $$\boldsymbol{\varepsilon}(\mathbf{v}) = \frac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^T)$$

- We can express the bilinear form in terms of the gradient of the test function:

$$B(\mathbf{u}, \mathbf{v}) = \int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{v}) \cdot \boldsymbol{L} \cdot \boldsymbol{\varepsilon}(\mathbf{u}), d\Omega = \int_{\Omega} \left(\frac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^T) \right) : \boldsymbol{L} : \left(\frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right.$$

Function Spaces for Test & Trial Functions

To ensure our weak form is mathematically well-posed, the solution (trial) function $\boldsymbol{u}$ and the test function $\boldsymbol{v}$ can't be just any function. They must belong to specific function spaces.

Why do we need restrictions? The bilinear form **B(u,v)** involves integrals of derivatives of the functions. For these integrals to be well-defined and finite, the functions must have a certain degree of smoothness.

Hilbert-Sobolev Spaces ($H^1(\Omega)$ and $H_0^1(\Omega)$:

A function u belongs to H^1 if both the function and its first derivatives are square-integrable over the domain Ω.

$H^1(\Omega) = \left(\mathbf{u} \in L^2(\Omega); |; \nabla \mathbf{u} \in L^2(\Omega)\right)$

In other words:

$$\int_\Omega \nabla \boldsymbol{u} \cdot \nabla \boldsymbol{u}\, d\Omega + \int_\Omega \boldsymbol{u} \cdot \boldsymbol{u}\, d\Omega < \infty$$

This ensures that the energy norm, is finite.

Handling Dirichlet Conditions $H_0^1(\Omega)$:

The test functions $\boldsymbol{v}$ must also respect the homogeneous Dirichlet boundary conditions. This leads us to the space $H_0^1(\Omega)$, which is a subspace of $H^1(\Omega)$.

It contains all functions that are zero on the Dirichlet boundary Γ_u.

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H^1_0(\Omega) = \left{ \mathbf{v} \in H^1(\Omega) ;\middle|; \mathbf{v} = \mathbf{0
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Our Problem Restated:

Find the solution $\boldsymbol{u} \in H^1(\Omega)$ such that $\boldsymbol{u} = \boldsymbol{u}^*$ on the Dirichlet boundary Γ_u and for all test functions $\boldsymbol{v} \in H_0^1(\Omega)$, the weak form holds:

$B(\boldsymbol{u}, \boldsymbol{v}) = F(\boldsymbol{v})$

Non-Zero Displacements & Function Spaces

- **Question**: If our prescribed displacement $\boldsymbol{u}^*$ on the boundary Γ_u is not zero, how does that affect our function spaces? Does the solution $\boldsymbol{u}$ still belong to $H_0^1(\Omega)$?
- **Answer**: No, the solution $\boldsymbol{u}$ will not belong to $H_0^1(\Omega)$ if $\boldsymbol{u}^*$ is non-zero on Γ_u.
- **The Distinction**: We must distinguish between the space for the solution (the trial space) and the space for the test function (the test space).

- Test Function $\boldsymbol{v}$: The test function v always belongs to $H_0^1(\Omega)$, which means it is zero on the Dirichlet boundary Γ_u. It represents an arbitrary variation or perturbation of the solution, and to keep the solution fixed at the boundary, the perturbation there must be zero.
- Trial Function $\boldsymbol{u}$: The solution $\boldsymbol{u}$ belongs to a different space, which we denote as $H^1(\Omega)$, but it is not constrained to be zero on the boundary Γ_u. It is a space containing all admissible functions that satisfy the essential boundary condition.

  The "Lifting" Technique: Mathematically, we handle this by splitting the solution u into two parts:

  $\mathbf{u} = \mathbf{u}_0 + \mathbf{u}_D$

where $\boldsymbol{u}_D$ is a known function (aka “lifting function”) that satisfies the Dirichlet boundary condition on Γ_u, and $\boldsymbol{u}_0$ is the unknown part of the solution that we seek.

Modified Weak Form:

- We substitute this decomposition into our weak form:
  $B(\mathbf{u}_0 + \mathbf{u}_D, \mathbf{v}) = F(\mathbf{v})$
- Using the linearity of B(· ,v):
  $B(\mathbf{u}_0, \mathbf{v}) + B(\mathbf{u}_D, \mathbf{v}) = F(\mathbf{v})$
- We then move all the known quantities to the right-hand side:
  $B(\mathbf{u}_0, \mathbf{v}) = F(\mathbf{v}) - B(\mathbf{u}_D, \mathbf{v})$
- Our new problem is to find the homogeneous component $\boldsymbol{u}_0 \in H_0^1(\Omega)$ that satisfies this modified equation for all

test functions $\boldsymbol{v} \in H_0^1(\Omega)$:
$B(\mathbf{u}_0, \mathbf{v}) = F(\mathbf{v}) - B(\mathbf{u}_D, \mathbf{v})$

- This means we are looking for a solution $\boldsymbol{u}_0$ that satisfies the weak form with respect to the homogeneous space $H_0^1(\Omega)$, while $\boldsymbol{u}_D$ is already known and satisfies the Dirichlet boundary condition.
- Once $\boldsymbol{u}_0$ is found, the final solution is simply $\boldsymbol{u} = \boldsymbol{u}_0 + \boldsymbol{u}_D$. In practice, this is handled during the assembly of the finite element system.

Principle of Minimum Potential Energy

An alternative but equivalent way to derive the weak form for elasticity problems is by minimizing a potential energy functional. This principle states that the exact solution to an elastic problem is the one that minimizes the total potential energy.

Total Potential Energy Π: The total potential energy of the system is the sum of the internal strain energy and the potential energy of the external loads.

$$\Pi(\mathbf{u}) = \underbrace{\frac{1}{2}\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{u}), d\Omega}_{\text{Strain Energy}} - \underbrace{\left(\int \Omega \mathbf{f} \cdot \mathbf{u}, d\Omega + \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{u}, d\Gamma \right)}_{\text{Potential of External Loads}}$$

Finding the Minimum: The solution u that minimizes the functional Π is found by setting its directional derivative to zero for any arbitrary perturbation (test function) v.
This is analogous to setting the first derivative to zero in standard calculus.

$$\lim_{\epsilon \to 0} \frac{\Pi(\mathbf{u} + \epsilon \mathbf{v}) - \Pi(\mathbf{u})}{\epsilon} = 0$$

Recovering the Weak Form: Calculating this derivative (which is the first variation of the functional) yields:

$$\delta\Pi = \int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}), d\Omega - \int_{\Omega} \mathbf{f} \cdot \mathbf{v}, d\Omega - \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{v}, d\Gamma = 0$$

Rearranging this gives us our familiar weak form:

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}), d\Omega = \int_{\Omega} \mathbf{f} \cdot \mathbf{v}, d\Omega + \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{v}, d\Gamma$$

This shows that the solution satisfying the weak form is also the one that minimizes the total potential energy of the system.

The Weak Form Solution Minimizes Energy

We previously showed that minimizing the potential energy functional leads to the weak form. Now we will prove the converse: the unique solution u to the weak form is the unique minimizer of the potential energy functional $L(\mathbf{w})$

- Define the Energy Functional:

$$L(\mathbf{w}) \stackrel{\text{def}}{=} \frac{1}{2}B(\mathbf{w},\mathbf{w}) - F(\mathbf{w})$$

- The Goal: We want to show that J(u)≤J(w) for any other admissible function $\boldsymbol{w}$ (a functions that satisfies the same Dirichlet boundary conditions as $\boldsymbol{u}$).

- Let's evaluate the energy of any admissible function w: We can write any such w as the sum of the true solution u and an error function e, so w=u+e.
- Since u and w both satisfy the same Dirichlet boundary conditions, the error function e must be zero on the Dirichlet boundary Γ_u and thus belongs to the space $H_0^1(\Omega)$, which is the space of functions that are square-integrable and have square-integrable first derivatives, and are zero on the Dirichlet boundary.
- Substitute and Expand: Let's substitute w=u+e into the energy functional:

$$L(\mathbf{w}) = L(\mathbf{u}+\mathbf{e}) = \frac{1}{2}B(\mathbf{u}+\mathbf{e},\mathbf{u}+\mathbf{e}) - F(\mathbf{u}+\mathbf{e})$$

- Using the linearity of the forms, this expands to:

$$L(\mathbf{w}) = \frac{1}{2}(B(\mathbf{u},\mathbf{u}) + 2B(\mathbf{u},\mathbf{e}) + B(\mathbf{e},\mathbf{e})) - (F(\mathbf{u}) + F(\mathbf{e}))$$

- Group the Terms: Now, we rearrange the expression:

$$L(\mathbf{w}) = \underbrace{\left(\frac{1}{2}B(\mathbf{u},\mathbf{u}) - F(\mathbf{u})\right)}_{L(\mathbf{u})} + \underbrace{(B(\mathbf{u},\mathbf{e}) - F(\mathbf{e}))}_{=0} + \underbrace{\frac{1}{2}B(\mathbf{e},\mathbf{e})}_{\frac{1}{2}|\mathbf{e}|_E^2}$$

- The middle term is zero because u is the true solution, so it satisfies the weak form B(u,v)=F(v) for any test function v, including our error function e.

- The Final Result: This simplifies to a fundamental relationship:

$$L(\mathbf{w}) = L(\mathbf{u}) + \frac{1}{2}|\mathbf{w} - \mathbf{u}|_E^2$$

- Since the energy norm $||w - u||_E^2$ is always greater than or equal to zero, this proves that L(w)≥L(u).
- The energy of any admissible function is greater than or equal to the energy of the true solution. The minimum is achieved only when $||w - u||_E^2 = 0$ which means $w = u$.

FEM Approximation & Error Estimates

- The Finite Element Method does not find the exact solution , but rather an approximate solution u_h . **The accuracy of this approximation is a central concern.**
- Discretization: We divide our domain Ω into smaller elements of characteristic size h. We then approximate the solution using simple polynomial functions (e.g., linear, quadratic) on each element.
- Error Measurement: The error is the difference between the exact solution u and the finite element solution u_h

- We often measure this error using the energy norm:
  $||\mathbf{e}||_E^2 = ||\mathbf{u} - \mathbf{u}_h||_E^2 = B(\mathbf{u} - \mathbf{u}_h, \mathbf{u} - \mathbf{u}_h)$
  This norm is physically meaningful as it represents the strain energy of the error.

- A Priori Error Estimate: For well-posed problems, we can estimate the error before solving. The error in the energy norm is bounded by a constant C (independent of the mesh) times terms involving the element size h and the polynomial order p of the shape functions.
  $||\mathbf{u} - \mathbf{u}_h||_E^2 \leq C^2(u, p)h^{2\gamma}$
  Where $\gamma = \min(r - 1, p)$, with r being the regularity of the solution and p being the polynomial order of the finite element shape functions.

- Key Takeaway: The error decreases as the mesh is refined (as h gets smaller) or as the polynomial order of the elements (p) is increased.
- We can compare the solutions for two different meshes or polynomial orders to estimate the convergence rate of the method. (next lecture)

FEM implementation - Discretize the Domain

The first step in any Finite Element analysis is to move from the continuous domain Ω to a discrete one. We replace the single integral over the whole domain with a sum of integrals over small, simple subdomains called elements. Our weak form is:

$$\underbrace{\int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{v})^T \mathbf{L} \boldsymbol{\varepsilon}(\mathbf{u}), d\Omega}_{\text{Bilinear Form } B(\mathbf{u},\mathbf{v})} = \underbrace{\int \Omega \mathbf{v}^T \mathbf{f}, d\Omega + \int_{\Gamma_t} \mathbf{v}^T \bar{\mathbf{t}}, d\Gamma}_{\text{Linear Form } F(\mathbf{v})}$$

Note: We've switched to transpose notation for Voigt vectors, common in implementation.

We approximate the domain Ω as a mesh of N_{el} finite elements, each with its own subdomain Ω_e

The integral over Ω becomes a sum of integrals over each element Ω_e

The bilinear and linear forms are now expressed as sums:

$$B(\mathbf{u}, \mathbf{v}) = \sum_{e=1}^{N_{el}} \int_{\Omega_e} \boldsymbol{\varepsilon}(\mathbf{v})^T \mathbf{L} \boldsymbol{\varepsilon}(\mathbf{u}), d\Omega_e$$

$$F(\mathbf{v}) = \sum_{e=1}^{N_{el}} \left(\int_{\Omega_e} \mathbf{v}^T \mathbf{f}, d\Omega_e + \int_{\Gamma_{t,e}} \mathbf{v}^T \bar{\mathbf{t}}, d\Gamma_e \right)$$

Where $\Gamma_{t,e}$ is the part of the element's boundary that lies on the traction boundary Γ_t

FEM implementation - Approximate Fields with Shape Functions

Within each element, we approximate the continuous displacement field using simple polynomial functions (shape functions) and the displacement values at the element's nodes.

The displacement u(x) at any point inside an element is interpolated from the element's nodal displacements d_e using shape functions: .

$$\mathbf{u}(\mathbf{x}) \approx \mathbf{u}_h(\mathbf{x}) = \mathbf{N}(\mathbf{x})\mathbf{d}_e$$

N(x) is the shape function matrix. For a 3D problem, it relates the 3 displacement components to the nodal values. For an element with n nodes, it's a 3×3n matrix:

$$\mathbf{N} = \begin{bmatrix} \phi_1 & 0 & 0 & \phi_2 & 0 & 0 & \dots & \phi_n & 0 & 0 \\ 0 & \phi_1 & 0 & 0 & \phi_2 & 0 & \dots & 0 & \phi_n & 0 \\ 0 & 0 & \phi_1 & 0 & 0 & \phi_2 & \dots & 0 & 0 & \phi_n \end{bmatrix}$$

where $\phi_i(x)$ is the scalar shape function for node i.

We do exactly the same for the test function v, relating it to a vector of arbitrary nodal values c e ：

$$\mathbf{v}(\mathbf{x}) \approx \mathbf{v}_h(\mathbf{x}) = \mathbf{N}(\mathbf{x})\mathbf{c}_e$$

FEM implementation - Compute Strains with the B-Matrix

With the displacement field approximated, we can find the strain by taking its derivatives. This process creates the crucial strain-displacement matrix, B.
The strain vector $\boldsymbol{\varepsilon}$ is found by applying a differential operator matrix, ∂, to the displacement vector $\boldsymbol{u}$.

$$\boldsymbol{\varepsilon} = \partial \mathbf{u} \quad \text{where for 3D engineering strain,} \quad \partial = \begin{pmatrix} \partial_x & 0 & 0 \\ 0 & \partial_y & 0 \\ 0 & 0 & \partial_z \\ 0 & \partial_z & \partial_y \\ \partial_z & 0 & \partial_x \\ \partial_y & \partial_x & 0 \end{pmatrix}$$

$$\boldsymbol{\varepsilon}_h = \partial\,(\mathbf{N}\mathbf{d}_e) = (\partial \mathbf{N})\,\mathbf{d}_e$$

We define the strain-displacement matrix B as the derivatives of the shape function matrix:

$$\mathbf{B} := \partial \mathbf{N}$$

This gives the direct algebraic relationship between strain and nodal displacements for an element:

$$\boldsymbol{\varepsilon}_h = \mathbf{B}\mathbf{d}_e$$

The B matrix contains the spatial derivatives of the shape

functions and is a function of the position within the element.

FEM implementation - The B-Matrix from the Symmetric Gradient

The crucial B-matrix is not arbitrary; it arises directly from the fundamental definition of the small strain tensor, ε, as the symmetric part of the displacement gradient.

1. Start with the Strain Definition:

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right)$$

2. Form the Voigt Strain Vector: For implementation, we use the Voigt notation with engineering shear strains γ_{ij} (where $\gamma_{ij} = \varepsilon_{ij} + \varepsilon_{ji}$, so $\gamma_{ij} = 2\varepsilon_{ij}$ for symmetric strains,

$\boldsymbol{\varepsilon}\text{Voigt} = \begin{pmatrix} \varepsilon xx & \varepsilon_{yy} & \varepsilon_{zz} & \gamma_{yz} & \gamma_{xz} & \gamma_{xy} \end{pmatrix} = \begin{pmatrix} & lt; 3 > partialu/\partial x & \partial v/\partial y & \partial w/\partial z & ($

3. Define the Differential Operator: We can express this relationship using a differential operator matrix, ∂, that acts on the displacement vector:

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\boldsymbol{\varepsilon}{\text{Voigt}} = \underbrace{ &lt;4>begin{pmatrix} \partial_x
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This explicitly shows how the structure of ∂ is determined by the physics of strain.

4. Introduce the FE Approximation: Finally, substitute the approximation for the displacement field, $u_h = Nd_e$, into the strain definition:

$\boldsymbol{\varepsilon}_h = \partial(\mathbf{N}\mathbf{d}_e) = (\partial\mathbf{N})\mathbf{d}_e$

This gives us our definition of the B-matrix, B:=∂N, and the final element-level relationship:

$$\boldsymbol{\varepsilon}_h = \mathbf{B}\mathbf{d}_e$$

FEM implementation - The Algebraic System:

$$k_e d_e = f_e$$

By substituting the shape function approximations for u and v into the element integral, we transform the differential equation into a system of linear algebraic equations.

Consider the bilinear form for one element, and substitute $\boldsymbol{\varepsilon}(u) = \boldsymbol{B}\boldsymbol{d}_e$ and $\boldsymbol{\varepsilon}(v) = \boldsymbol{B}\boldsymbol{c}_e$

$$\int_{\Omega_e} (\mathbf{B}\mathbf{c}_e)^T \mathbf{L}(\mathbf{B}\mathbf{d}_e), d\Omega_e = \mathbf{c}_e^T \left(\int \Omega_e \mathbf{B}^T \mathbf{L}\mathbf{B}, d\Omega_e \right) \mathbf{d}_e$$

We define the term in the parentheses as the element stiffness matrix k_e

$$\mathbf{k}_e := \int \Omega_e \mathbf{B}^T \mathbf{L} \mathbf{B}, d\Omega_e$$

Similarly, for the linear form, we get the element force vector f_e:

$$\mathbf{f}_e := \int \Omega_e \mathbf{N}^T \mathbf{f}, d\Omega_e + \int_{\Gamma_{t,e}} \mathbf{N}^T \bar{\mathbf{t}}, d\Gamma_e$$

The weak form for the element must hold for any arbitrary c_e, this can only be true if:

$$\mathbf{k}_e \mathbf{d}_e = \mathbf{f}_e$$

FEM implementation - Numerical Integration

The integrals for k_e and f_e involve the B and N matrices, which can be complex polynomials. These are almost always computed numerically using Gaussian Quadrature. The idea is to approximate an integral by a weighted sum of the integrand's value at specific sample points, known as Gauss points.

First, we map the element Ω_e to a simple reference element

The integral transforms as:

$$\int_{\Omega_e} g(\mathbf{x}), d\Omega_e = \int_{\hat{\Omega}} g(\mathbf{x}(\boldsymbol{\xi})), |\mathbf{J}(\boldsymbol{\xi})|, d\hat{\Omega}$$

where |J| is the determinant of the Jacobian of the coordinate mapping.

The integral over the reference element is then approximated by the sum:

$$\int_{\hat{\Omega}} g(\tilde{\boldsymbol{\xi}}), d\hat{\Omega} \approx \sum_{i=1}^{N_{gp}} w_i g(\tilde{\boldsymbol{\xi}}_i)$$

where w_i are the Gauss weights and $\boldsymbol{\xi}_i$ are the Gauss point coordinates.

Applying this to our stiffness matrix:

$$\mathbf{k}_e = \int \Omega_e \mathbf{B}^T \mathbf{L} \mathbf{B}, d\Omega \approx \sum_{i=1}^{N_{gp}} \left(\mathbf{B}(\boldsymbol{\xi}_i)^T \mathbf{L} \mathbf{B}(\boldsymbol{\xi}_i) \right) |\mathbf{J}(\boldsymbol{\xi}_i)| w_i$$

The force vector f_e is computed in the same way:

$$\mathbf{f}_e = \int \Omega_e \mathbf{N}^T \mathbf{f}, d\Omega_e + \int_{\Gamma_{t,e}} \mathbf{N}^T \mathbf{t}^*, d\Gamma_e \approx \sum_{i=1}^{N_{gp}} \left(\mathbf{N}(\boldsymbol{\xi}_i)^T \mathbf{f}(\boldsymbol{\xi}_i) + \mathbf{N}(\boldsymbol{\xi}_i)^T \mathbf{t}^*(\boldsymbol{\xi}_i) \right) |\mathbf{J}(\xi$$

FEM implementation - Algorithmic overview

1. Initialize: Create empty global stiffness matrix K and force vector F.
2. Loop over Elements: For each element e in the mesh…
3. Initialize Element Matrices: Create empty local

stiffness matrix k_e and force vector f_e.

4. Loop over Gauss Points: For each Gauss point i in the element…
5. Calculate at Point: Evaluate the B matrix and the Jacobian determinant |J| at this point's coordinates.
6. Accumulate Integral: Compute the integrand for

k_e and f_e. Multiply by the Gauss weight and |J| and add to the element matrices.

7. Assemble: Once the loops over Gauss points are done, add the completed element matrices k_e and f_e into the global K and F based on the element's node numbering.

8. Apply Boundary Conditions: Modify the

global system $Kd = F$ to enforce the prescribed Dirichlet displacements.

9. Solve: Solve the final system of linear equations for the vector of all nodal displacements d.
10. Post-Process: Use the computed displacements d to find element strains and stresses. This is done using the same matrices B and L

as before at the gauss points.

1. K←0, F←0
2. for e = 1 to $N_e l$ do
3. $k_e \leftarrow 0, f_e \leftarrow 0$
4. for i = 1 to N_{gp} do
5. Calculate $\mathbf{B}(\boldsymbol{\xi}_i)$ and $|\mathbf{J}(\boldsymbol{\xi}_i)|$
6. k_e += $\mathbf{B}^T \mathbf{L} \mathbf{B}\, |\mathbf{J}(\boldsymbol{\xi}_i)|\, w_i$
   f_e += $\mathbf{N}^T \mathbf{f}\, |\mathbf{J}(\boldsymbol{\xi}_i)|\, w_i + \ldots$
7. Assemble
   $(\mathbf{K}, \mathbf{F}, \mathbf{k}_e, \mathbf{f}_e, \mathrm{dof_map}_e)$

8. end for

9. Modify K,F for $d_j = d^*$

10. Solve $\mathbf{Kd} = \mathbf{F}$ for $\mathbf{d}$

11. end for

12. $\boldsymbol{\sigma}_e = \boldsymbol{L}\boldsymbol{B}$\oldsymbol$d_e$,

$\boldsymbol{\varepsilon}_e = \boldsymbol{B}\mathbf{d}_e$

Input: Mesh (Ω_e), Material Properties ($\mathbf{L}$), Body Forces ($\mathbf{f}$), Boundary Conditions ($\bar{\mathbf{u}}, \bar{\mathbf{t}}$)

Output: Nodal Displacements ($\mathbf{d}$), Element Stresses ($\boldsymbol{\sigma}$), Element Strains ($\boldsymbol{\varepsilon}$)

Initialize Global Stiffness $\mathbf{K} \leftarrow \mathbf{0}$ and Force $\mathbf{F} \leftarrow \mathbf{0}$.

For each element e in the mesh:

a. Initialize Element Stiffness $\mathbf{k}_e \leftarrow \mathbf{0}$ and Force $\mathbf{f}_e \leftarrow \mathbf{0}$.

b. For each Gauss point i in element e:

- Compute $\mathbf{B}(\boldsymbol{\xi}_i)$, Jacobian determinant $|\mathbf{J}(\boldsymbol{\xi}_i)|$, and weight w_i.
- Accumulate stiffness: $\mathbf{k}_e \mathrel{+}= \mathbf{B}^T\mathbf{L}\mathbf{B}, |\mathbf{J}|w_i$
- Accumulate body forces: $\mathbf{f}_e \mathrel{+}= \mathbf{N}^T\mathbf{f}, |\mathbf{J}|w_i$

c. Assemble traction boundary forces into $\mathbf{f}_e$ if applicable.

d. Assemble local into global: Assemble$(\mathbf{K}, \mathbf{F}, \mathbf{k}_e, \mathbf{f}_e)$.

Apply Dirichlet boundary conditions to the global system $(\mathbf{K}, \mathbf{F})$.

Solve the linear system $\mathbf{Kd} = \mathbf{F}$ for the displacement vector $\mathbf{d}$.

For each element e in the mesh (Post-processing):

a. Extract element displacements d_e from global d.

b. Compute strains $\boldsymbol{\varepsilon}_e = \mathbf{B}d_e$ and stresses $\boldsymbol{\sigma}_e = \mathbf{L}\boldsymbol{\varepsilon}_e$ at desired points.

Speaker notes